ABES Engineering College, Ghaziabad

Department of Applied Sciences & Humanities

Session: 2025-26 **Semester: II**

Course Code: 25AS202

Course Name: Applied Calculus and Differential Equations **Max. Marks: 10**

Assignment 1

Date of Assignment: /02/26 **Date of submission: /03/26**

S.No.	KL	CO	Question	Marks
1.	K3	CO4	Solve the differential equation $\frac{d^2 i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = 0$ where $R^2 C = 4L$ and R, C, L are constants.	2
2.	K3	CO4	Solve the differential equation $[4D^2 + 4D - 3]y = e^{2t}$.	2
3.	K3	CO4	Find the general solution (complete solution) of $(D-2)^2 y = 8(e^{2x} + \sin 2x + x^2)$.	2
4.	K3	CO4	Solve : $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = xe^{3x} + \text{Sin } 2x$.	2
5.	K3	CO4	Solve the differential equation $\frac{d^4y}{dx^4} - y = \cos x \cosh x$.	2
6.	K3	CO4	Solve: $x^3\frac{d^3y}{dx^3} + 3x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = x + \log x$	2
7.	K3	CO4	Solve by changing the independent variable $x^6\frac{d^2y}{dx^2} + 3x^5\frac{dy}{dx} + a^2 y = \frac{1}{x^2}$	2
8.	K3	CO4	Apply method of variation of parameter, solve: $\frac{d^2y}{dx^2} + y = \text{cosec } x$	2
9.	K3	CO4	Using variation of parameter method solve: $x^2 y'' + xy' - y = x^3 e^x$.	2
10.	K3	CO4	An R-L-C circuit connected in a series has R = 90Ω , C = $\frac{1}{140}$ farad, L =10 henries and an applied voltage E(t) = 10 Cos t . Assuming no initial charge on the capacitor but an initial current of 1 ampere at t =0, when the voltage is first applied, find the subsequent charge on the capacitor.	2

Answers:

1. $i = (c_1 + c_2 t)e^{-\frac{R}{2L}t}$

2. $y = c_1 e^{0.5\,t} + c_2 e^{-1.5\,t} + \frac{1}{21}e^{2t}$

3. $y = [c_1 + c_2 x]e^{2x} + 4x^2 e^{2x} + \cos 2x + 2x^2 + 4x + 3$

4. $y = c_1 e^{x} + c_2 e^{2x} - \frac{e^{3x}}{2}\left(x - \frac{3}{2}\right) + \frac{1}{20}(3\cos 2x - \sin 2x)$

5. $y = c_1 e^{x} + c_2 e^{-x} + c_3\, cosx + c_4\, sinx - \frac{1}{5}\cos x \cosh x$

6. $y = \frac{c_1}{x} + \sqrt{x}\left\{c_2 \,\mathrm{Cos}\left(\frac{\sqrt{3}}{2}\log x\right) + c_3 \,\mathrm{Sin}\left(\frac{\sqrt{3}}{2}\log x\right)\right\} + \frac{x}{2} + \log x$

7. $y = c_1 \cos\left(\frac{a}{2x^2}\right) + c_2 \sin\left(\frac{a}{2x^2}\right) + \frac{1}{a^2x^2}$

8. $y = c_1\, cosx + c_2\, sinx - x\, cosx + sinx.\, log(sinx)$

9. $y = c_1 x + \frac{c_2}{x} + \left(x - 3 + \frac{3}{x}\right)e^{x}$

10. $q = \frac{3}{25}e^{-2t} - \frac{43}{250}\, e^{-7t} + \frac{1}{250}\,(9\, sint + 13\, cost)$